Automatic Graph Learning Convolutional Networks for Hyperspectral Image Classification

Jie Chen, Licheng Jiao, *Fellow, IEEE*, Xu Liu, *Member, IEEE*, Lingling Li, *Member, IEEE*, Fang Liu, *Senior Member, IEEE*, and Shuyuan Yang, *Senior Member, IEEE*

***Abstract*— The excellent performance of graph convolutional networks (GCNs) on non-Euclidean data has drawn widespread attention from the hyperspectral image classification (HSIC) community, where the predefined graph (including node modeling and adjacency matrix calculation) plays a key role. However, existing GCN-based methods rely on manual efforts in constructing and updating graphs, and the superpixel-based node features lack high-level semantics. In this article, we propose an automatic graph learning convolutional network (Auto-GCN), which unifies the graph learning and HSIC in a "network-in-network" manner. Specifically, the graph is employed to model the interaction of the high-order tensors. Considering the powerful learning and representation capabilities of convolutional neural networks (CNNs), the semisupervised Siamese network (SiamNet) is embedded into GCNs and HSIC networks to accomplish the automatic learning and dynamic updating of the graph. GCNs further encode and infer the dynamic graph, and then, the learnable graph reprojection matrix is designed to assign graph representations to pixels. The dynamic graph serves the HSIC task during forward propagation, while the HSIC task continuously corrects the graph during backward propagation. Therefore, the "automatic" of the proposed Auto-GCN is not only reflected in the fact that the graph representation is designed and updated by an end-to-end network but is also HSIC task-oriented. The experimental results show that the proposed Auto-GCN outperforms other state-of-the-art methods on four publicly available hyperspectral datasets.**

***Index Terms*— Automatic learning, dynamic graph, graph convolutional network (GCN), hyperspectral image classification (HSIC), Siamese network (SiamNet).**

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China, International Research Center of Intelligent Perception and Computation, School of Artificial Intelligence, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: chenjie818826@163.com; lchjiao@mail.xidian.edu.cn).

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I. Introduction

OWING to the interaction between physical components, objects can absorb and reflect electromagnetic radiation of specific wavelengths. Hyperspectral imaging techniques, a revolution in remote sensing, can record the spectral response of various materials in hundreds of continuous and subdivided high-resolution spectral bands. Hence, hyperspectral images (HSIs) contain abundant spectral and spatial information of ground objects [1]. HSI classification (HSIC) aims to classify scenes in a pixel-by-pixel manner and has been widely used in various fields, such as geology, ecological monitoring, urban land planning, and military defense [2], [3].

In earlier years, models such as support vector machines (SVMs), random forest (RF), and sparse learning (SL) are broadly adopted for HSIC [4]–[6]. Such traditional machine learning algorithms extremely rely on handcrafted features. With the research of neuroscience, brain science, and cognitive science, neural networks simulate the complex hierarchical cognition of the human brain by increasing the number of network layers and backpropagation [7]. In the background of early theoretical research and the Big Data era, the concept of deep learning [8] was proposed, which combines feature extraction, feature selection, and classifier design in an end-to-end framework. The breakthrough of deep CNNs in image recognition dataset ImageNet leads to the rapid adoption of deep learning methods for computer vision [9], [10]. Nowadays, CNNs are the primary method for recognition and detection tasks [11]–[13]. Furthermore, the graph convolutional networks (GCNs) [14] extend the convolution operation to the non-Euclidean domain (graphs and manifolds). In the field of image recognition and detection, there are increasing attempts to encode the dependencies and message passing between objects with GCNs [15]–[19].

The HSI data have a highly nonlinear distribution structure, and abundant spectra will cause different objects to share similar spectral features [20], meaning that the "same object with the different spectrum, the same spectrum with different object." Besides, spectra are pixel-based features that are insensitive to orientation and scale shift. How to fully utilize the spatial and spectral information of hyperspectral data to extract the high-level semantic information is a long-standing problem for HSI interpretation. Several state-of-the-art methods validate the effectiveness of global contextual information

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modeling for improving the performance of semantic segmentation. Some methods expand the receptive field by stacking convolution and pooling layers [21], [22], and some capture the remote dependencies by computing the interactions between arbitrary locations [23]–[25]. However, CNNs enrich the context by stacking semantic information and do not perform information transfer and inference processes. Moreover, CNNs generate convolutional features only along with spatial coordinates while ignoring the topological information of images. Image is a regular grid data, while the linkage of the graph is free, which can provide a geometric framework for knowledge association and message transmission [26]. The superiority of graph-structured data lies not only in its focus on the relationships between nodes but also in its universal existence. Therefore, we apply the recently proposed GCNs [14] to model the superior semantics of HSI: global topological information.

Recently, a group of GCN-based HSIC algorithms have emerged and realized excellent performance [27]–[29]. For instance, Mou *et al.* [30] construct a graph structure based on the original HSI, i.e., a hyperspectral pixel point represents a node, and the edge is the similarity of any two hyperspectral pixels. Liu *et al.* [31] merge the irregular superpixel level features encoded by GCNs and the regular region features encoded by CNNs, allowing CNNs and GCNs to collaborate in a single network. However, the fixed initial graph is more sensitive to noise, which will lead to inaccurate edge weights of the nodes [32]. Therefore, many works update the graph structure while adjusting the feature embedding. In the proposed multiscale dynamic graph [33], the dynamic update of the graph is realized by fusing the current data with the graph in the previous layer, and the multiscale information is achieved by constructing graphs with different neighborhood scales. The context-aware dynamic GCN (CAD-GCN) [34] performs different distance metrics for each graph convolution layer to generate adaptive graph edge weights. Jiang *et al.* [35] construct hypergraphs for the high-order features of each layer in the deep neural network; specifically, the k-NN method is applied to generate the hyperedges, and then, the hyperedges are extended by the k-means clustering algorithm. Liu *et al.* [36] dynamically update the structural relationships of potential samples by adding the extra manifold regularization term to the original target function of the GCNs.

Although existing GCN-based algorithms have made various degrees of progress, there are still some limitations.

1) Graph nodes are typically represented by pixels or superpixels, which would create hundreds of thousands or even millions of nodes [30], [31]. This will induce excessive computational effort. Moreover, node features are commonly represented by pixel coordinates or raw spectra [33]. Such shallow features lack robustness and are susceptible to the intraclass variation and interclass similarity of HSI.
2) The distance function (e.g., the Euclidean distance and the Mahalanobis distance) is frequently applied to calculate the adjacency matrix [27], [32]. However, the manually defined distance function is weaker in fitting the high-dimensional nonlinear HSI data [37].
3) The design of dynamic graphs relies on manual fusion and updating and does not take the requirements of the HSIC task into consideration [33], [34].

To solve the problems above, in this article, a novel automatic graph learning convolutional network (Auto-GCN) is proposed, which builds a graph based on the high-order tensors of image regions, and then, the semisupervised dual-input Siamese network (SiamNet) [38] is employed to compute the similarity between high-dimensional node features. In addition, the graph is learned and updated during the training process to empower the task adaptation of the graph representations. The proposed Auto-GCN is trained end-to-end. To sum up, the main contributions of this article are given as follows.

1) A novel Auto-GCN is proposed for the HSIC task, where the design and learning of the graph are all performed by neural networks.
2) A semisupervised SiamNet for constructing the higher order tensor graph is proposed to improve the first two points of the aforementioned limitations of existing methods. During the training of SiamNet, a novel sample labeling approach based on the IoU metric is proposed to relabel the HSI.
3) A new dynamic graph update method as the gradient of HSIC loss decreases is proposed. SiamNet, GCNs, and the hyperspectral classification network are jointly trained to produce more meaningful graph representations than the manually computed dynamic graph.

The rest of this article is organized as follows. Section II introduces the related work. Section III describes the proposed Auto-GCN in detail. Section IV validates and analyzes the proposed method on four public datasets. Section V concludes this article and the future directions.

II. Related Work

A. HSI Classification

Since the experience-based feature extraction methods are poor in describing the essential properties of high-dimensional nonlinear hyperspectral data, a large number of CNNs-based algorithms have emerged in the HSIC community. Chen *et al.* [39] first propose a deep learning HSIC method. Furthermore, the birth of fully convolutional networks (FCNs) [40] starts the pixel-to-pixel semantic segmentation pattern. The large-scale deep networks effectively avoid the limitations of patch-by-patch feature learning.

For the complex and variable hyperspectral data structure, the effective use of spatial and spectral information remains a challenge for many researchers to overcome. Numerous works construct spatial–spectral models to improve the performance of HSIC. For instance, Zhang *et al.* [41] extract multiple feature representations by probabilistic SVMs, and the extended MRF models are then applied to combine multiple features with spatial constraints. Cao *et al.* [42] use MRF to obtain the spatial information between neighboring pixels and the SVM classifier to obtain the spectral information, which is then merged into the energy function of MRF. There are

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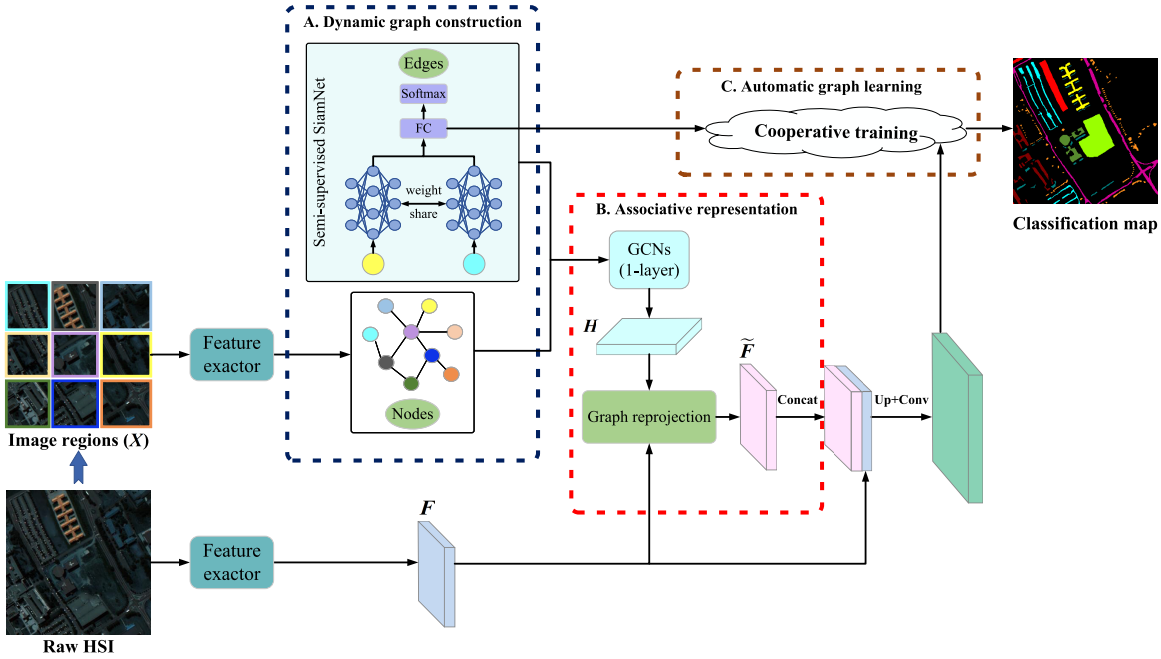


Fig. 1. Pipeline of the proposed Auto-GCN.

also plenty of deep learning spatial–spectral methods. For instance, Zhao *et al.* [43] capture the spectral and spatial features of HSI separately with the dual-channel CNNs architecture. Feng *et al.* [44] propose the spatial–spectral generating adversarial network (GAN) that designs two generators, a 1-D transposed convolutional network (1-D-TCN) and a 2-D-TCN, to generate spectra and spatial patches, respectively. Then, the discriminator extracts joint spatial–spectral features and outputs multiclass probabilities by 1-D CNNs and 2-D CNNs.

To get the discriminative representations, some methods concern the exploitation of the geometric information of HSI. Zhang *et al.* [45] perform the joint representation of six shape-flexible rectangular regions around a pixel (respectively, the central region, the original region, and four orientation-oriented regions). The proposed interleaving perception CNNs [46] maintain the complementary structure of multisource data with the idea of second-order interaction alignment of the Gram matrix. Hao *et al.* [47] add the recurrent neural networks (RNNs) to the head of CNNs to simulate the human brain and construct a geometry-aware ResNet loss based on the pretrained ResNet. Moreover, Ge *et al.* [48] present the respective feature- and decision-level fusion methods based on residual networks to enhance the discrimination and stability of the model.

B. GCNs Meet Remote Sensing

Graph formalism is ubiquitous throughout the natural and social sciences, and graph-based methods are broadly used in remote sensing image processing. Research has shown that manifold learning and graph regularization help to reveal the intrinsic geometry and distinguishing structure of the data space [49]. Thus, some work leverages graph regularization techniques to preserve the structural information of the data [50]–[52]. There are also approaches that treat pixel semantics as random variables and then use the probabilistic graphical model for postprocessing of the semantic segmentation task [53], [54].

The proposal of the graph neural networks (GNNs) [55] brings graph representation learning to a new stage. Shuman *et al.* [56] not only give the definition of graph Fourier transform but also clarify the concepts of filtering, convolution, and modulation of graph signal processing, which establishes the mathematical foundation for the graph convolution. Furthermore, GCNs generalize convolution operations to the non-Euclidean domain and are grouped into two categories: spectral domains and spatial domains (also called vertex domains). Spatial methods define "local" connection and pooling layers by analogy to the compactly supported filters, and stride convolutions and pooling in CNNs [57], [58]. The spectral methods extend the definition of convolution in the Fourier domain to general graphs using the graph Laplacian matrix [14], [59]–[61]. To express more complex correlations between data, Feng *et al.* [62] generalize GCNs to hypergraph domains using the hypergraph Laplacian matrix. Today, GCNs have proven to be highly competitive in areas, such as social networks [63], molecular biology [64], natural language processing [65], and emotion recognitionn [66].

Also, a quantity of GCN-based works has emerged in the remote sensing community. For instance, Du *et al.* [67] construct the multimodel graph adjacency matrix with the weight matrix that fuses the multisource remote sensing data. Fu *et al.* [68] model the graph structure information in both feature space and example space. Liu *et al.* [69] encode the global contextual information through GCNs. There are also some works using GCNs for the semantic segmentation of 3-D point cloud data [70]–[72].

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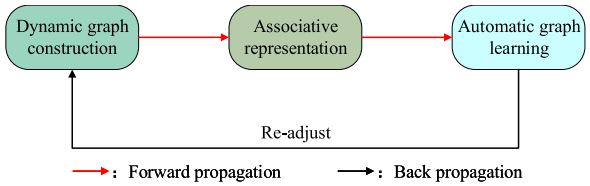


Fig. 2. Relevance of the three parts of the proposed Auto-GCN.

III. Methodology

Grounded in the perceptual organization theory, graph inference of image content has drawn much attention, where image regions are graph nodes and similarities between regions are served as edges. GCNs are then performed to capture remote dependencies between semantics and encode the topological information. However, in graph modeling, the node features lack abstract representations and the weights of edges fail to match the high-dimensional data characteristics. Moreover, the design of dynamic graphs often relies on calculation by hand. In this article, an automatic graph learning network is designed to solve these problems. During the training process, the dynamic graph is continuously learned and refined according to the statistical distribution and the classification task.

In this section, the pipeline of the proposed Auto-GCN is shown in Fig. 1. First, the row HSI is divided into grid-like regions. Section III-A describes the construction of the dynamic graph, where the region features are considered as nodes and the edges are learned by the weight-sharing SiamNet. In Section III-B, the dynamic graph is further encoded by one-layer GCNs and then completed the graph reprojection operation together with the whole HSI features to learn the graph-to-pixel assignment and, finally, get the associative representation of the whole HSI. After upsampling and convolution operations, the cascaded association representations and the HSI feature are fed to the classifier. Section III-C describes the automatic graph learning process where SiamNet, GCNs, and hyperspectral classification networks are trained collaboratively. In backpropagation, the dynamic graph is further readjusted according to the gradient descent. The correlation of these three parts is shown in Fig. 2.

Next, we detail the key components of Auto-GCN in the pipeline, which are Section III-A: dynamic graph construction, Section III-B: associative representation, and Section III-C: automatic graph learning.

A. Dynamic Graph Construction

Graph modeling is a fundamental and crucial part of applying graph representation learning to image processing. Graph representation learning generally consists of three steps: first, each node sends its own feature information to its neighbors; second, each node fuses the information from its neighbors; and third, after aggregating the previous information, a nonlinear transformation is performed to increase the representation capability of the model [73]. Therefore, node representations significantly affect the quality of graph-level representations. To obtain more abstract discriminative graph-level features, we embed SiamNet into GCNs and

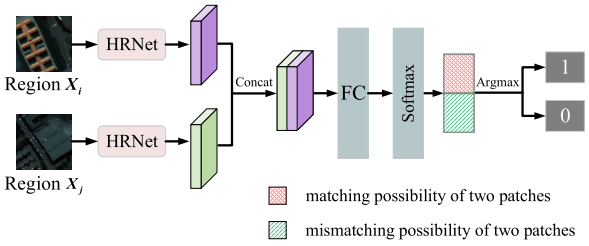


Fig. 3. Network structure of the SiamNet.

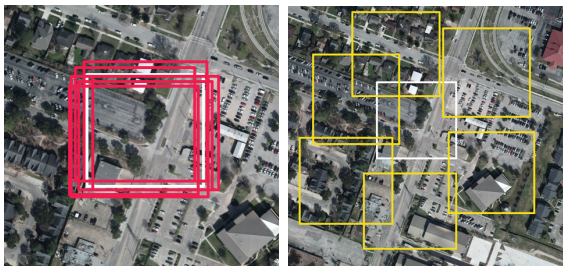


Fig. 4. Way to get the training samples for SiamNet. (a) Gauss. (b) Uniform.

HSIC to complete the construction of the dynamic graph in which rectangular regions are modeled as graph nodes. It is because node features based on superpixels are usually shallow features, such as coordinates and pixel mean values, which lack discriminativeness, moreover, rectangular regions are more convenient for extracting high-dimensional abstract node features using CNNs than irregular superpixel regions. The detailed procedure is given as follows.

As shown in Fig. 1, the whole HSI $\mathcal{X}$ is partitioned into regions $\boldsymbol{X} = [\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n]$ by the regular grid, and n is the number of regions

$$\boldsymbol{X}_i = \text{Grid}(\mathcal{X}; \alpha), \quad i = 1, \ldots, n \tag{1}$$

where α is the dividing size of the regions.

The image regions form the nodes of the dynamic graph, and then, deep CNNs are adopted to extract discriminative high-dimensional representations. The node features can be denoted as

$$\boldsymbol{N}_i = \Omega(\boldsymbol{X}_i) \tag{2}$$

where Ω indicates the feature extraction on the hyperspectral data. Note that all the feature extractors used in this article are the pretrained HRNet [74], [75].

Compared with the superpixel-based node modeling, the region graph not only models the interactions of the higher order features but also can capture longer range semantic dependencies with the same number of convolutional layers of GCNs.

Based on the viewpoint of manifolds, the data that we can observe are the unfolding of low-dimensional manifolds in a high-dimensional space [76]. However, calculating the similarity between nodes by using a manually designed distance function ignores the statistical properties and intrinsic structure of high-dimensional data. To avoid this problem, SiamNet is

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used to evaluate the edge weights. The "Siamese" of the neural network is realized by sharing the weights. It aims to learn a similarity measure from the data for comparing and matching samples with unknown categories.

The input of SiamNet is $(\boldsymbol{X}_i, \boldsymbol{X}_j, \boldsymbol{Y})$, where $(\boldsymbol{X}_i, \boldsymbol{X}_j)$ denotes a region pair and $(\boldsymbol{X}_i, \boldsymbol{X}_j) \in \boldsymbol{X}$. The input label $\boldsymbol{Y}$ is 0 or 1: 0 means that the input data belong to the same category and 1 means a different category. To reduce the computational overhead, we perform binary classification using an off-line trained SiamNet, and the architecture is shown in Fig. 3. The parameter shared deep convolutional network in SiamNet has the same structure as the feature extractor Ω. The output tensor of the two-branch SiamNet is cascaded and expanded into a 1-D vector and then is fed into the fully connected (FC) and softmax layers for similarity learning. The loss function is defined as

$$\mathcal{L}_{\text{siam}} = -\frac{1}{m}\sum_{i=1}^{m}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{3}$$

where m is the number of training samples, y_i is the input label, and $\hat{y}_i$ is the probability of the predicted label being 1, which is calculated as

$$\hat{y}_i = \log\left(\frac{e^{r_1}}{e^{r_0}+e^{r_1}}\right) \tag{4}$$

where r_0 and r_1 are the two output values of the FC layer, which are nonnegative and sum to 1, indicating the matching or mismatching possibility of the two patches.

Since the classification labels of the original data are not applicable to SiamNet for similarity learning, we design a new labeling framework to generate positive and negative training samples, as shown in Fig. 4. Fig. 4(a) shows the generation process of positive samples, and Fig. 4(b) shows the negative samples. In the whole HSI, the anchor boxes (the white box in Fig. 4) are generated by scratching the windows with overlap. In the training process, we take positive and negative samples around the anchor boxes in a ratio of 1:4. Positive and negative samples are assigned with the Intersection over Union (IoU) value as the measuring standard. IoU is the result of dividing the overlap area by the union area.

Positive Sample: Generate boxes with $0.8 < \text{IoU} < 1$ around the anchor box in the Gaussian mode, such as the red boxes in Fig. 4(a). The white and red boxes are randomly grouped together as the input pairs of SiamNet and labeled with 0.

Negative Sample: Generate boxes with $0 < \text{IoU} < 0.5$ around the anchor box in the uniform and global mode, such as the yellow boxes in Fig. 4(b). The white and yellow boxes are randomly grouped together as the input pairs of SiamNet and labeled with 1.

B. Associative Representation

Given the constructed graph, in this section, we describe the specific steps for getting the association representation, including the graph encoding of GCNs and the design of the learnable graph reprojection matrix.

Graphs are non-Euclidean data, which consist of nodes and edges connecting the nodes, can be formulated as $\mathcal{G} = (\mathcal{V}, \varepsilon, \boldsymbol{A})$, where $\mathcal{V}$ is the set of nodes and ε is the set of edges. $\boldsymbol{A}$ is the weight adjacency matrix, a real symmetric matrix. The adjacency matrix is a convenient way to represent the graph in which the elements represent the interactions between nodes. $\boldsymbol{A}_{ij}$ denotes the weight of the edge (i, j) between vertices i and j. $\boldsymbol{A}_{ij} = 0$ means that there is no edge between i and j. The most commonly used definition of $\boldsymbol{A}$ is the threshold Gaussian kernel weight function

$$\boldsymbol{W}_{ij} = \begin{cases} \exp\left(-\frac{[\text{dist}(i,j)^2]}{2\gamma^2}\right), & \text{dist}(i,j) \le k \\ 0, & \text{others} \end{cases} \tag{5}$$

where γ is a parameter that controls the smoothness of the Gaussian kernel function and $\text{dist}(i, j)$ can be either the physical distance between vertices or the distance between features. The simplest $\boldsymbol{A}_{ij} \in \{0, 1\}$ and $\boldsymbol{A}_{ij} = 1$ when $(i, j) \in \varepsilon$; otherwise, $\boldsymbol{A}_{ij} = 0$.

The node degree calculates the number of node's neighbors, which is the most obvious and direct node feature. The degree of node i is $d_i = \sum_{j\in\mathcal{V}} \boldsymbol{A}_{ij}$. The degree matrix $\boldsymbol{D}$ is a diagonal matrix with size $N \times N$ (N denotes the number of nodes with $|\mathcal{V}| = N$), and the diagonal elements are the degree of each node. Analogously to the Laplace operator on the lattice, the graph Laplace is defined as $\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{A}$. To produce unbiased estimates, the Laplace matrix is usually normalized by $\boldsymbol{L}_{\text{norm}} = \boldsymbol{D}^{-(1/2)}(\boldsymbol{D} - \boldsymbol{A})\boldsymbol{D}^{-(1/2)}$.

Since the eigenfunction of the Laplace operator is the same as the negative exponent of the frequency-domain mode in the Fourier transform, the Fourier transform can be generalized to arbitrary graphs by the eigendecomposition of the graph Laplace matrix $\boldsymbol{L} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^T$, where $\boldsymbol{\Lambda}$ is a diagonal matrix with diagonal elements as eigenvalues $\boldsymbol{\Lambda} = [\lambda_1, \ldots, \lambda_N]$. The eigenvalues of $\mathbf{L}$ behave similar to the frequencies in the Fourier transform. $\mathbf{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_N] \in \mathbb{R}^{N\times N}$ is the eigenvector matrix, defined as the Fourier mode on the graph. Then, the Fourier transform of the graph signal $\boldsymbol{f} \in \mathbb{R}^N$ can be expressed as $\hat{\boldsymbol{f}} = \boldsymbol{U}^T\boldsymbol{f}$, and the inverse Fourier transform is $\boldsymbol{f} = \boldsymbol{U}\hat{\boldsymbol{f}}$. The convolution of the signal $\boldsymbol{f}$ and filter h on the graph can be presented as

$$\begin{aligned} \boldsymbol{f} \star_{\mathcal{G}} h &= \boldsymbol{U}(\boldsymbol{U}^T\boldsymbol{f} \circ \boldsymbol{U}^T h) \\ &= \boldsymbol{U}(\boldsymbol{U}^T\boldsymbol{f} \circ \boldsymbol{\theta}_h) \\ &= (\boldsymbol{U}\text{diag}(\boldsymbol{\theta}_h)\boldsymbol{U}^T)\boldsymbol{f} \end{aligned} \tag{6}$$

where $\star_{\mathcal{G}}$ indicates that the convolution is specific to the graph $\mathcal{G}$, $\boldsymbol{\theta}_h = \mathbf{U}^T h \in \mathbb{R}^{|V|}$, and $\text{diag}(\boldsymbol{\theta}_h)$ is the diagonal matrix. Furthermore, the Laplace eigenvalues are used to parameterize $\boldsymbol{\theta}_h$, and the spectrum filter is defined as the polynomial function of the matrix Λ, which is the r-degree polynomial of the eigenvalues of $\mathbf{L}$. Then,

$$\boldsymbol{f} \star_{\mathcal{G}} h = (\mathbf{U}\mathcal{P}_r(\boldsymbol{\Lambda})\mathbf{U}^T)\boldsymbol{f} = \mathcal{P}_r(\boldsymbol{L})\boldsymbol{f}. \tag{7}$$

Estrach *et al.* [77] first give the definition of the nonparametric spectral filter [see (6)] and the parametric spectral filter [see (7)]; the latter depends on the r-hop neighborhoods of the nodes (i.e., the neighbors that can be reached by walking r steps) and ensures the local properties of the convolution. ChebNet [60] defines $\mathcal{P}_r(\boldsymbol{L})$ with the Chebyshev polynomial. The kth-order Chebyshev polynomial can be expressed as

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$T_k(\boldsymbol{x}) = 2\boldsymbol{x}T_{k-1}(\boldsymbol{x}) - T_{k-2}(\boldsymbol{x})$, where $T_0 = 1$ and $T_1 = \boldsymbol{x}$. Thus, the filter can be parameterized into a truncated form $h_{\boldsymbol{\theta}}(\boldsymbol{L}) = \sum_{k=0}^{K-1} \boldsymbol{\theta}_k T_k(\widetilde{\boldsymbol{L}})$, where the parameter $\boldsymbol{\theta} \in \mathbb{R}^K$ is the coefficient vector of Chebyshev. $T_k(\widetilde{\boldsymbol{L}})$ is the kth-order Chebyshev polynomial at the scaled Laplacian matrix $\widetilde{\boldsymbol{L}} = 2\boldsymbol{L}/\lambda_{\max} - \boldsymbol{I}$. Applying the Chebyshev polynomial to the convolution transform yields

$$\boldsymbol{f} \star_{\mathcal{G}} h = \sum_{k=0}^{K-1} \boldsymbol{\theta}_k T_k(\widetilde{\mathbf{L}})\boldsymbol{f}. \tag{8}$$

Kipf and Welling [14] define the basic graph convolution layers, which restricts the layer convolution operations to local first-order approximations (i.e., first-order Chebyshev polynomial approximations) and renormalize the normalized Laplace matrix to mitigate exploding/vanishing gradients in multilayer models. The convolutional output of the lth layer is represented as

$$\boldsymbol{H}^{(l)} = \sigma\left(\widetilde{\boldsymbol{A}}\boldsymbol{H}^{(l-1)}\boldsymbol{W}^{(l)}\right) \tag{9}$$

where $\widetilde{\boldsymbol{A}} = (\boldsymbol{D}+\boldsymbol{I})^{-(1/2)}(\boldsymbol{I}+\boldsymbol{A})(\boldsymbol{D}+\boldsymbol{I})^{-(1/2)}$ is a normalized variant of the adjacency matrix, $\boldsymbol{W}^{(l)}$ is a learnable parameter matrix, and $\sigma(\cdot)$ denotes the activation function. Based on this model, we propose a fresh adjacency matrix calculation method driven by neural networks.

As shown in Fig. 5, to measure the similarity between any pair of vertices, pairs of nodes are sent to the trained SiamNet, and the output $\hat{y}$ (the probability of the predicted label being 1 in SiamNet) is taken as the edge weights. Substituting this operation into (9), we have the dynamic graph convolution

$$\boldsymbol{H}^{(l)} = \sigma\left(\widetilde{\boldsymbol{\Phi}}(\boldsymbol{X};\boldsymbol{W}_f)\boldsymbol{H}^{(l-1)}\boldsymbol{W}_g^{(l)}\right) \tag{10}$$

where $\boldsymbol{\Phi}$ denotes the adjacency matrix calculated by SiamNet and $\widetilde{\boldsymbol{\Phi}}$ is the normalized form of it. $\boldsymbol{W}_f$ is the parameters of the FC layer in the off-line trained SiamNet, and $\boldsymbol{W}_g^{(l)}$ is the trainable parameter matrixes of the lth layer of GCNs. The elements of $\boldsymbol{\Phi}$ can be calculated as

$$\boldsymbol{\Phi}_{ij} = \hat{y} = \text{SiamNet}(\boldsymbol{N}_i, \boldsymbol{N}_j), \quad (i, j) \in 1, 2, \ldots, n. \tag{11}$$

In this article, one-layer GCN is used and the output can be expressed as

$$\boldsymbol{H} = \text{Relu}(\widetilde{\boldsymbol{\Phi}}(\boldsymbol{X};\boldsymbol{W}_f)\boldsymbol{N}\boldsymbol{W}_g). \tag{12}$$

Inspired by the work of Li and Gupta [78], we construct the learnable graph reprojection matrix to invert the graph representations into grid features (which we name as associative representations). As shown in the below branch of Fig. 1, the whole HSI $\mathcal{X}$ is fed into the feature extractor to get the feature map $\boldsymbol{F}$ of size $h \times w \times d_1$

$$\boldsymbol{F} = \Omega(\mathcal{X}). \tag{13}$$

Then, the features $\boldsymbol{F}$ and $\boldsymbol{H}$ are cooperated to perform the graph reprojection operation. The graph reprojection matrix $\boldsymbol{Q} \in \mathbb{R}^{hw\times n}$ is constructed according to the feature map $\boldsymbol{J} \in \mathbb{R}^{hw\times d_1}$, which is obtained by reshaping $\boldsymbol{F}$, where n is the number of image regions. Meanwhile, to increase the learnability of the reprojection operation, we introduce two

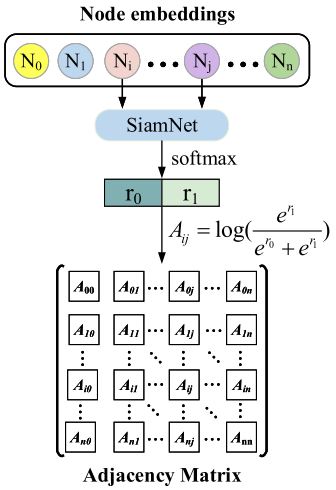


Fig. 5. Calculation of the adjacency matrix.

parametric matrixes: $\boldsymbol{W}^b \in \mathbb{R}^{d_1\times n}$ and $\boldsymbol{Var} \in \mathbb{R}^{d_1\times n}$. Then, the column vector $\boldsymbol{q}_i$ of $\boldsymbol{Q}$ can be calculated as

$$\boldsymbol{q}_i = \frac{\exp\left(-\left\|\left(\boldsymbol{J} - \boldsymbol{W}_i^b\right)/\boldsymbol{Var}_i\right\|_2^2/2\right)}{\sum_{hw}\exp\left(-\left\|\left(\boldsymbol{J} - \boldsymbol{W}_i^b\right)/\boldsymbol{Var}_i\right\|_2^2/2\right)}. \tag{14}$$

Thus, the associative representations $\widetilde{\boldsymbol{F}}$ can be expressed as

$$\widetilde{\boldsymbol{F}} = \boldsymbol{Q}\boldsymbol{H}^T \tag{15}$$

where $\widetilde{\boldsymbol{F}} \in \mathbb{R}^{hw\times d_2}$, and d_2 is the dimension of the output layer in GCNs.

C. Automatic Graph Learning

In Sections III-A and III-B, dynamic graph construction and the calculation of the association representations are introduced, respectively. Then, we cascade the feature $\boldsymbol{F}$ and the associative representations $\widetilde{\boldsymbol{F}}$ to obtain the stronger discriminative features, which are sent to the upsampling and convolution layers to perform multitask learning. The output feature map $\boldsymbol{O}$ is denoted as

$$\boldsymbol{O} = \text{Up_Conv}(\boldsymbol{F} \parallel \widetilde{\boldsymbol{F}}) \tag{16}$$

where $\parallel$ means the cascade operation of the feature dimension, and the output $\boldsymbol{O}$ is of the same size as the raw HSI. The loss function of the Auto-GCN is

$$\begin{aligned}\mathcal{L}_{\text{AG}} &= -\frac{1}{P}\sum_{i=1}^{P}\sum_{c=1}^{C}\boldsymbol{W}_c\boldsymbol{G}_i^c\boldsymbol{O}_i^c \\ &= -\frac{1}{P}\sum_{i=1}^{P}\sum_{c=1}^{C}\boldsymbol{W}_c\boldsymbol{G}_i^c\left(\log\frac{\exp(a_i^c)}{\sum_{j=1}^{C}\exp(a_i^j)}\right)\end{aligned} \tag{17}$$

where C is the category number of the HSI, P is the pixels number of the whole HSI, and a_i is the probability that pixel point i is predicted to be class c. $\boldsymbol{W}_c, \boldsymbol{G}_C \in \mathbb{R}^C$, and $\boldsymbol{W}_c$ denote the weight of categories, and $\boldsymbol{G}_i^c$ is the one-hot vector of the ground truth.

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To obtain more adaptive association representations for classification tasks and hyperspectral data, we perform collaborative training for SiamNet, GCNs, and HSIC. Thus, the dynamic graph can be further updated by using the gradient descent algorithm. To improve the training efficiency of the classifier, we fix the convolutional layer parameters of SiamNet and only update the FC layer parameters during the Auto-GCN training. The detailed training procedure of the proposed Auto-GCN is shown in Algorithm 1.

Algorithm 1 Learning Process of the Proposed Auto-GCN

Input: A whole HSI $\mathcal{X}$, feature extractor $\mathbf{\Omega}$, iteration of SiamNet $iter_{siam}$, iteration of Auto-GCN $iter_{AG}$, the parameters of the SiamNet during the offline training $\boldsymbol{W}_s$, the parameters of FC layer in the offline trained SiamNet $\boldsymbol{W}_f$, the parameters of GCNs $\boldsymbol{W}_g$, the parameters matrix $\boldsymbol{Var}$ and $\boldsymbol{W}^b$ in graph reprojection, the remaining trainable parameters of Auto-GCN $\boldsymbol{W}_m$;

1) ***Preparation***
   a) Divide the whole HSI $\mathcal{X}$ into image regions by regular grid;
   b) Obtain the training samples of SiamNet in the manner of Section III-A;
2) ***Step 1: Offline training of SiamNet***
   a) Optimizing *min* $\mathcal{L}_{siam}$, Updating $\boldsymbol{W}_s$;
   b) Repeat the training process for $iter_{siam}$ times;
3) ***Step 2: Associative representation***
   a) Get the node embedding (Eq. 2) and the adjacency matrix (Eq. 11), and then feed them into the 1-layer GCNs (Eq. 12).
   b) Calculate the graph reprojection matrix (Eq. 14) and get the associative representations (Eq. 15).
4) ***Step 3: Automatic graph learning***
   a) Feed the cascade feature map (Eq. 16) into the classifier;
   b) Co-training SiamNet, GCNs and classification networks and optimizing *min* $\mathcal{L}_{AG}$, the parameters $\boldsymbol{W}_m$, $\boldsymbol{W}_f$, $\boldsymbol{W}_g$, $\boldsymbol{Var}$ and $\boldsymbol{W}^b$ are updated;
   c) Repeat the training process for $iter_{AG}$ times;

Output:
Classification result.

IV. Experimental Results

In this section, we conduct sufficient experiments on three public HSI datasets[1] to validate the effectiveness of the proposed Auto-GCN.

A. Data Description

1) Indian Pines Hyperspectral Dataset: The scene is located at the Indian Pine test site in Northwest Indiana, USA, and was collected by the AVIRIS sensor in 1992. The dataset contains 16 ground cover categories with size of 145 × 145 pixels and the spatial resolution of 20 m × 20 m. By removing the bands covering the absorption region, the dataset retains 200 spectral reflectance bands for analysis with wavelengths ranging from 400 to 2500 nm.

2) Pavia University Hyperspectral Dataset: This scene was acquired by the Reflection Optical System Imaging Spectrometer (ROSIS) sensor at the University of Pavia, Italy. The scene has a geometric resolution of 1.3 m, contains nine ground classes, and consists of 610 × 340 pixels and 103 bands for analysis, covering the spectral range from 430 to 860 nm.

3) Salinas Hyperspectral Dataset: The data were acquired by the AVIRIS sensor in the Salinas Valley, CA, USA. The scene contains 16 ground categories with a size of 512 × 217 pixels and a spatial resolution of 3.7 m × 3.7 m. After removing the 20 absorption bands from the salt marsh scene, 204 spectral bands were used for the analysis.

4) Houston University Hyperspectral Dataset: The dataset was acquired on the University of Houston campus and adjacent urban areas, and was used in the 2013 IEEE GRSS Data Fusion Contest. The image size is 349 × 1905 pixels and has 144 spectral bands in the 380–1050-nm range. The scene contains 15 land cover classes with a spatial resolution of 2.5 m × 2.5 m.

For the four HSI datasets mentioned above, we randomly select 30 training samples for each category, and 15 samples

TABLE I
Network Structure of the Proposed Auto-GCN

	Operation	Output shape
Feature extractor	Input HSI	(B_1,H_1, W_1, S)
	HRNetV2 [2]	(B_1, H_1/4, W_1/4, d_1)
SiamNet head	Input patches	(B_2, H_2, W_2, S)
	Concat (dim = 3)	(B_2, H_2/4, W_2/4, $2d_1$)
	View	(B_2, H_2/4 × W_2/4 × $2d_1$)
	FC	(B_2, 2)
Auto-GCN head	HSI feature	(B_1, H_1/4, W_1/4, d_1)
	Patches feature	(n, H_2/4, W_2/4, d_1)
	1-layer GCNs	(n, d_2)
	Graph reprojection	(H_1/4 × W_1/4, d_2)
	Reshape + Repeat	(B_1, H_1/4, W_1/4, d_2)
	Concat (dim = 3)	(B_1, H_1/4, W_1/4, d_1 + d_2)
	Upsample	(B_1, H_1/2, W_1/2, d_1 + d_2)
	Conv 1×1	(B_1, H_1/2, W_1/2, 64)
	BN	(B_1, H_1/2, W_1/2, 64)
	Relu	(B_1, H_1/2, W_1/2, 64)
	Dropout	(B_1, H_1/2, W_1/2, 64)
	Upsample	(B_1, H_1, W_1, 64)
	Conv 1×1	(B_1, H_1, W_1, C)
	Output	(B_1, H_1, W_1, C)

TABLE II
Accuracy of SiamNet on Each Dataset

Data Sets	IP	PU	Sa	HU
Accuracy	99.62	99.35	99.85	99.78

[1]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[2]https://github.com/leoxiaobin/deep-high-resolution-net.pytorch

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are randomly selected for training if the total sample of the categories is less than 30. In the experimental records, we denote the Indian Pines dataset by "IP," the Pavia University dataset by "PU," the Salinas dataset by "Sa," and the Houston University dataset by "HU."

B. Experimental Settings

1) Implementation Details: All the networks in this article are implemented on the Pytorch platform. The equipment used in the following experiments consists of an Intel (R) Xeon (R) CPU E5-2630 v3 @ 2.40 GHz and one NVIDIA GeForce GTX TITAN X GPU with 12 GB of memory. The proposed method has two networks to be trained: SiamNet and Auto-GCN. The detailed network architecture configuration is shown in Table I, where SiamNet and Auto-GCN share the same feature extractor (HRNetV2). In Table I, S denotes the raw spectral dimension, *"B_1," "H_1,"* and *"W_1"* denote the batch size, height, and width of the HSI fed into Auto-GCN, and *"B_2," "H_2,"* and *"W_2"* are the batch size, height, and width of the image regions, respectively. *"d_1"* and *"d_2"* are the dimension of the last layer of HRNetV2 and one-layer GCNs, respectively, *"n"* denotes the node number, and *"C"* is the category number.

In the experiments, first, we perform the off-line training of SiamNet. The training samples of SiamNet are obtained in the manner of Section III-A, the learning rate of SiamNet is set to 0.001, the optimizer is SGD, the momentum is 0.9, the weight decay is 0.0005, and the epoch is set to 100. The off-line training accuracy of SiamNet for each dataset is shown in Table II. Then, to improve the convergence speed of Auto-GCN, we initialize the feature extractor of Auto-GCN with the parameters of the trained feature extractor in SiamNet. Finally, we follow the process of Algorithm 1 to complete the Auto-GCN training, where the Auto-GCN training includes fine-tuning the FC layer of the off-line-trained SiamNet. The epoch of Auto-GCN is 150, and we use the Adam optimizer with the base learning rate of 0.005, the momentum of 0.9, and the weight decay of 0.0005.

2) Evaluation Metrics: To quantitatively compare the classification performance of different models, we adopt four commonly used metrics, including per-class accuracy, OA, average accuracy (AA), and Kappa coefficient (Kappa) [79].

3) Comparison Algorithms: To evaluate the classification performance of the proposed Auto-GCN, eight state-of-the-art hyperspectral classification methods were also selected for comparison. Including four state-of-the-art GCN-based methods, they are minibatch GCN (Mini-GCN) [27], non-local GCN [30], CAD-GCN [34], multiscale dynamic GCN (MD-GCN) [33], and three CNNs' methods, which are deeper contextual CNN (DC-CNN) [80], spectral–spatial residual network (SSRN) [81], and deep FCN (DFCN) [82]. Note that the training sample number of the comparison algorithm is consistent with ours.

C. Sensitivity Analysis of Parameters

In the proposed Auto-GCN, there are several important hyperparameters that need to be selected manually. For example, to obtain the graph topology representation of HSI, Auto-GCN builds the graph based on image regions, so the region size affects the volume of information interaction between nodes. GCNs are used for graph coding, and therefore, it is also necessary to evaluate the impact of different spectral graph convolution models on the performance of Auto-GCN. As such, in this section, the hyperparameters that we evaluate in detail mainly include the selection of the image region size and the different spectral graph convolution models.

TABLE III

EFFECT OF THE IMAGE REGION SIZE ON THE KAPPA COEFFICIENT OF EACH DATASET AND THE NUMBER OF NODES CORRESPONDING TO DIFFERENT REGION SIZES (I.E., GRIDS ON HEIGHT × GRIDS ON WIDTH)

Size	IP	PU	Sa	HU
64	93.12±0.07	89.45±1.71	92.34±2.06	93.85±1.08
	3×3	10×5	8×4	6×30
96	**96.51±0.12**	88.39±0.49	97.04±1.62	93.90±0.39
	2×2	7×4	6×4	4×20
128	91.72±1.20	92.34±0.03	98.20±0.17	**95.73±0.41**
	2×2	5×3	4×2	3×15
160	-	**94.28±0.06**	**99.37±0.03**	95.18±0.04
	-	4×2	3×2	2×12
192	-	93.60±1.06	98.27±0.22	95.45±0.18
	-	3×2	3×2	2×10
224	-	92.81±0.92	-	95.14±0.02
	-	3×2	-	2×9
256	-	93.43±0.02	-	94.68±0.10
	-	3×2	-	2×8

TABLE IV

PERFORMANCE OF DIFFERENT SPECTRAL GRAPH CONVOLUTION MODELS ON THE PAVIA UNIVERSITY DATASET

ID	ChebNet		GCNs	
	hop=2	hop=3	1-layer	2-layer
1	92.35±0.59	91.77±0.18	**97.27±0.94**	93.51±0.17
2	99.86±0.02	99.87±0.02	97.14±0.39	**99.87±0.01**
3	94.61±0.48	**95.19±0.09**	92.50±0.34	93.93±0.06
4	89.81±0.71	89.54±0.60	87.25±1.49	**90.13±0.59**
5	**99.20±0.10**	99.07±0.17	98.28±0.86	99.06±0.13
6	99.44±0.04	99.40±0.03	**99.95±0.04**	99.50±0.08
7	98.67±0.16	98.25±0.45	**99.10±0.80**	98.54±0.28
8	77.63±0.32	77.45±0.25	**88.42±6.30**	78.08±0.30
9	96.17±0.29	96.09±0.09	93.16±4.78	**96.69±0.37**
OA	95.04±0.06	94.89±0.09	**95.55±0.11**	95.29±0.04
AA	94.19±0.07	94.07±0.11	**94.78±1.87**	94.37±0.11
Kappa	93.69±0.08	93.50±0.11	**94.28±0.06**	94.01±0.05

To measure the effect of region size on the classification accuracy, as shown in Table III, the optional region size is $\alpha = \{64, 96, 128, 160, 192, 224, 256\}$. We conduct a series of experiments on four datasets, and the kappas of different hyperparameters are separately recorded in Table III. Comprehensive to the experimental results, the region size of the Indian Pines dataset is set to 96, the Pavia University dataset is 160, the Salinas dataset is 160, and the Houston University dataset is 128, respectively.

Meanwhile, for choosing the proper spectral graph convolution model, we perform experiments on ChebNet [60] with

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TABLE V
NUMERICAL RESULTS ON THE INDIAN PINES DATASET

ID	Mini-GCN [27]	Nonlocal-GCN [30]	CAD-GCN [34]	MD-GCN [33]	DC-CNN [80]	SSRN [81]	DFCN [82]	GCN	GCN+LA	GCN+LA+DG
1	**100±0.00**	**100±0.00**	99.46±0.54	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	83.42±3.44	83.42±2.16	88.99±3.01
2	55.27±0.70	65.71±2.22	88.08±4.57	80.18±0.84	76.80±2.46	71.98±0.95	78.25±1.03	84.55±1.78	84.52±0.48	**97.04±0.58**
3	70.00±0.00	76.62±5.39	95.62±1.89	98.26±0.00	90.87±0.35	90.75±0.71	94.38±0.07	96.35±0.14	97.21±0.30	**98.71±0.46**
4	81.80±0.91	88.21±0.20	97.85±2.75	**98.57±0.00**	88.32±0.08	87.21±1.20	88.41±0.34	91.08±0.78	90.38±2.79	95.97±0.65
5	54.08±0.65	94.19±1.46	93.79±3.71	95.14±0.33	90.73±1.56	93.67±1.04	90.07±0.02	91.42±1.47	91.32±1.33	**95.22±0.25**
6	85.43±0.62	95.33±1.75	96.41±1.78	97.16±0.57	98.52±0.47	97.86±0.00	**98.86±0.45**	96.66±0.10	96.25±0.68	94.03±0.16
7	92.31±0.00	92.31±0.00	97.95±2.04	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	94.53±3.74	94.53±0.74	**100±0.00**
8	**100±0.00**	**100±0.00**	99.81±0.05	98.89±0.00	**100±0.00**	**100±0.00**	**100±0.00**	99.90±0.04	99.76±0.13	95.71±0.18
9	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	**100±0.00**	95.04±2.02	91.36±1.22	97.00±1.76
10	65.75±1.00	77.25±1.57	89.34±4.03	90.02±1.02	86.41±2.05	82.66±0.90	90.34±0.40	93.13±0.17	94.77±1.05	**97.42±0.10**
11	48.05±4.26	78.06±0.00	93.09±3.14	93.35±1.47	93.97±0.88	92.40±0.61	95.71±0.08	96.76±1.39	96.57±1.60	**98.91±0.15**
12	70.81±1.48	79.57±0.00	93.41±2.54	93.05±2.30	79.04±0.75	77.98±0.06	82.24±1.01	89.02±0.01	91.79±0.55	**99.13±0.23**
13	71.43±0.00	97.71±0.00	99.79±0.19	**100±0.00**	97.71±0.00	97.71±0.13	97.71±0.74	87.19±1.89	88.46±0.08	80.36±0.43
14	90.26±0.31	92.58±0.85	99.36±0.54	**99.72±0.05**	98.46±0.59	97.65±0.21	99.68±0.66	99.18±0.70	99.42±0.37	99.70±0.19
15	66.85±0.00	98.22±1.06	99.20±0.57	99.02±0.06	99.34±0.00	99.63±0.00	**99.72±0.03**	99.14±0.42	98.77±0.95	96.05±0.40
16	**100±0.00**	**100±0.00**	98.91±1.01	95.71±0.00	**100±0.00**	**100±0.00**	**100±0.00**	87.56±1.59	90.13±0.96	89.54±1.73
OA	66.84±1.12	82.51±1.10	94.13±0.78	93.47±0.38	90.92±0.08	89.40±0.35	92.55±0.36	93.88±1.40	94.22±1.44	**96.98±0.10**
AA	78.25±0.38	89.75±0.72	**96.38±0.35**	96.24±0.21	93.79±0.32	93.17±0.12	95.97±0.93	92.81±1.12	93.04±0.29	95.24±0.09
Kappa	62.53±1.24	80.16±1.24	93.29±0.88	92.55±0.43	89.61±0.91	87.88±0.39	91.56±0.08	92.96±1.05	93.36±1.60	**96.51±0.12**

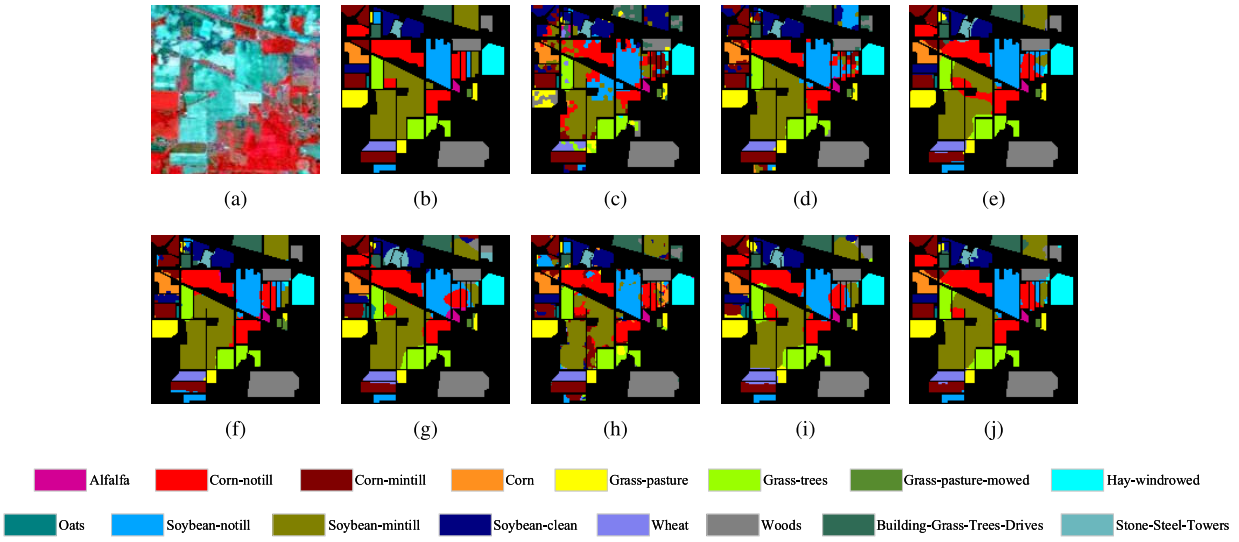


Fig. 6. Visualization results on the Indian Pines dataset. (a) False Color. (b) GT. (c) Mini-GCN. (d) Nonlocal-GCN. (e) CAD-GCN. (f) MD-GCN. (g) DC-CNN. (h) SSRN. (i) DFCN. (j) Ours.

TABLE VI
NUMERICAL RESULTS ON THE PAVIA UNIVERSITY DATASET

ID	Mini-GCN [27]	Nonlocal-GCN [30]	CAD-GCN [34]	MD-GCN [33]	DC-CNN [80]	SSRN [81]	DFCN [82]	GCN	GCN+LA	GCN+LA+DG
1	78.55±0.22	92.2±0.00	83.85±4.80	93.55±0.37	94.92±1.60	96.32±1.59	94.58±1.28	91.52±3.36	90.39±0.49	**97.27±0.94**
2	85.33±1.07	81.06±0.13	95.56±1.97	99.25±0.23	73.61±3.66	81.74±0.39	69.88±0.17	93.54±2.68	**99.74±0.03**	97.14±2.39
3	86.88±0.05	75.39±0.09	**95.50±2.41**	92.03±0.24	48.09±1.27	85.90±2.64	47.96±1.08	91.34±3.95	91.20±1.27	92.50±0.34
4	88.44±0.23	95.84±0.08	81.90±3.51	83.78±1.55	96.35±0.38	96.43±0.57	**96.74±0.92**	87.95±0.23	85.96±0.29	87.25±1.49
5	98.00±0.33	97.14±0.18	98.91±1.06	99.47±0.09	97.14±0.00	97.14±0.00	97.14±0.00	98.86±1.01	**99.03±0.43**	98.28±0.86
6	75.55±1.10	99.79±0.10	97.81±2.89	95.26±0.50	91.37±6.62	99.83±0.17	90.02±0.47	99.90±0.02	99.94±0.02	**99.95±0.04**
7	92.78±0.49	98.75±0.00	96.79±2.42	98.92±1.04	86.51±12.45	99.84±0.00	86.51±1.45	98.81±0.77	98.55±0.51	**99.10±0.80**
8	65.78±2.13	92.54±0.35	95.35±2.76	94.99±1.33	98.26±0.25	95.28±1.89	**98.36±0.14**	82.91±2.86	80.65±1.71	88.42±6.30
9	**100±0.00**	96.21±0.11	84.80±5.33	81.03±0.49	96.32±0.00	95.12±1.23	96.33±0.06	84.95±1.15	82.49±1.20	93.16±4.78
OA	85.70±1.52	88.11±0.04	92.91±1.01	**95.68±0.22**	83.11±2.55	89.88±0.71	81.28±4.44	91.98±1.77	94.10±0.28	95.55±0.11
AA	85.70±1.03	92.11±0.03	92.27±0.94	93.15±0.28	86.96±2.43	94.32±0.77	86.39±3.06	92.20±1.40	91.99±0.22	**94.78±1.87**
Kappa	83.92±1.73	84.76±0.05	90.69±1.29	94.25±0.29	78.66±1.10	87.00±0.91	76.57±5.23	89.80±2.33	92.52±0.36	**94.28±0.06**

different hops and GCNs [14] with different layer counts. ChebNet [see (8)] approximates the convolution kernel using K-order Chebyshev polynomials, which has strong expressiveness despite having the high complexity and parametric number. GCNs [see (9)] only cover the first-order neighbor nodes and expand the perceptual field by stacking multiple graph convolution layers. GCNs can be trained faster and have more practicality than ChebNet. The experimental results of

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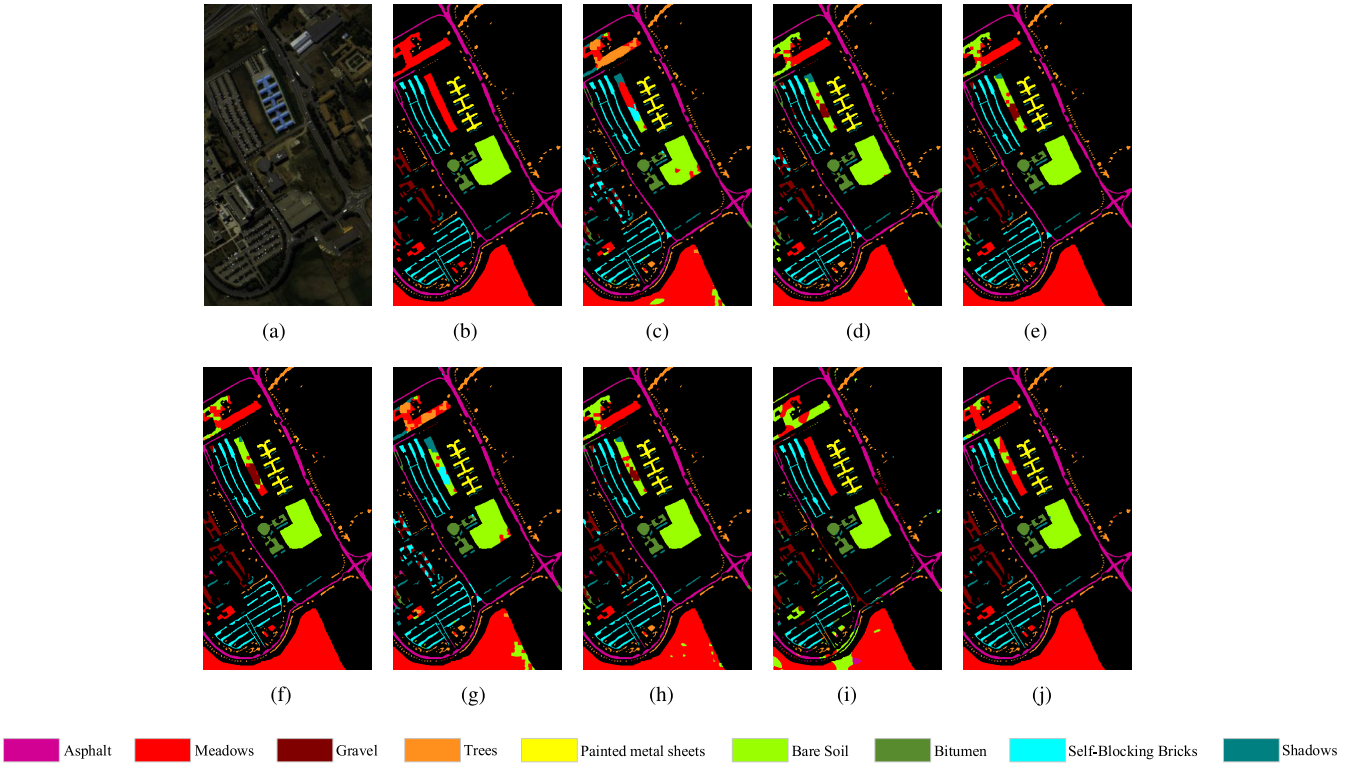


Fig. 7. Visualization results on the Pavia University dataset. (a) False Color. (b) GT. (c) Mini-GCN. (d) Nonlocal-GCN. (e) CAD-GCN. (f) MD-GCN. (g) DC-CNN. (h) SSRN. (i) DFCN. (j) Ours.

TABLE VII
NUMERICAL RESULTS ON THE SALINAS DATASET

ID	Mini-GCN [27]	Nonlocal-GCN [30]	CAD-GCN [34]	MD-GCN [33]	DC-CNN [80]	SSRN [81]	DFCN [82]	GCN	GCN+LA	GCN+LA+DG
1	96.61±2.54	99.96±0.00	**100±0.00**	**100±0.00**	71.93±9.76	99.47±0.05	**100±0.00**	99.99±0.01	99.98±0.02	**100±0.00**
2	74.39±0.93	86.69±3.12	**100±0.00**	99.91±0.09	86.69±1.24	99.73±0.19	99.63±0.29	99.86±0.19	99.80±0.20	**100±0.00**
3	98.72±0.51	99.17±0.02	**99.97±0.02**	96.49±2.44	99.17±0.11	98.85±1.56	99.05±0.27	99.01±0.20	99.89±0.06	99.39±0.14
4	83.23±3.04	91.12±0.19	**99.55±0.08**	93.92±3.99	91.12±1.94	85.96±1.24	87.13±7.57	84.31±5.95	95.32±4.24	98.58±0.11
5	97.93±0.58	97.95±0.10	98.46±1.05	98.46±0.33	97.95±0.07	93.87±5.84	96.02±2.80	95.79±2.66	97.66±0.33	**99.22±0.31**
6	98.13±1.20	98.11±0.06	97.77±0.33	99.13±0.36	98.11±0.74	98.81±0.26	99.02±0.56	99.28±0.49	99.01±0.38	**99.69±0.05**
7	96.93±1.87	95.45±0.22	99.61±1.56	99.90±0.01	95.45±4.32	93.61±6.02	97.83±2.05	97.11±2.60	98.84±1.08	**100±0.00**
8	96.06±1.97	72.56±2.31	**99.10±0.17**	98.60±1.19	72.05±1.19	82.76±9.86	92.75±5.63	97.40±2.03	96.36±3.25	98.75±0.07
9	99.17±0.02	99.96±0.01	95.80±0.00	98.77±1.01	**99.99±0.01**	99.95±0.01	99.92±0.04	99.94±0.05	99.98±0.02	99.85±0.03
10	95.83±3.07	92.83±0.20	**100±0.00**	97.25±1.98	88.50±7.80	94.72±0.37	95.78±1.88	97.52±2.23	97.07±2.61	99.25±0.00
11	**99.92±0.06**	97.10±0.02	98.01±0.27	98.60±0.35	97.10±1.76	98.96±0.03	99.15±0.30	98.93±0.53	98.61±0.33	99.54±0.42
12	96.25±0.80	95.89±0.17	99.82±0.09	99.61±0.21	95.89±4.05	99.05±0.22	99.04±0.20	99.00±0.24	98.83±0.06	**100±0.00**
13	86.71±6.86	93.68±0.09	**97.99±0.98**	93.97±1.25	93.68±2.87	93.04±3.78	94.81±1.28	94.19±1.13	94.79±0.92	99.95±0.04
14	92.40±3.74	95.38±0.45	97.89±1.53	96.52±0.18	95.47±1.85	95.02±2.29	95.51±1.60	96.10±2.15	97.53±0.90	**99.03±0.58**
15	99.07±0.31	99.57±0.01	97.47±1.65	95.36±2.92	**99.63±0.26**	99.48±0.18	98.98±0.53	97.04±2.51	96.45±2.90	99.42±0.18
16	89.11±5.89	88.44±0.11	99.76±0.20	99.44±0.56	88.44±4.44	95.33±4.03	99.08±0.62	99.39±0.60	99.26±0.61	**100±0.00**
OA	93.95±0.74	90.68±0.17	98.28±0.54	98.20±0.11	89.29±0.85	93.99±0.63	96.81±0.64	97.54±0.64	97.96±0.32	**99.43±0.03**
AA	93.78±0.38	93.99±2.53	98.78±0.22	97.93±0.34	91.95±1.09	95.54±2.28	97.11±1.24	97.19±1.28	98.09±0.12	**99.54±0.03**
Kappa	932±0.87	89.62±0.28	98.09±0.60	98.00±0.12	88.08±0.85	93.34±0.70	96.46±0.71	97.27±0.71	97.73±0.35	**99.37±0.03**

ChebNet and GCNs are recorded in Table IV. The table shows that GCNs perform better than ChebNet. Although the results of one-layer GCNs are comparable to two-layer GCNs, the former has fewer parameters, and therefore, we pick the one-layer GCNs' model.

D. Numerical and Visual Comparison With Other Algorithms

To evaluate the performance of the proposed Auto-GCN, we compared the classification results of Auto-GCN with other algorithms. Meanwhile, in order to investigate the effectiveness of each module in the proposed method, we run the model with different settings on four datasets. The quantitative results of the series of experiments on the Indian Pines, Pavia University, and Salinas datasets are recorded in Tables V–VIII, respectively, where the highest values in each row are marked in bold.

There are two major modules in the proposed Auto-GCN: one is the computation of adjacency matrix by SiamNet (that is, learnable adjacency matrix, denoted as "LA"), and the other

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TABLE VIII
NUMERICAL RESULTS ON THE HOUSTON UNIVERSITY DATASET

ID	Mini-GCN [27]	Nonlocal-GCN [30]	CAD-GCN [34]	MD-GCN [33]	DC-CNN [80]	SSRN [81]	DFCN [82]	GCN	GCN+LA	GCN+LA+DG
1	**100±0.00**	87.09±9.05	94.45±3.49	96.25±1.42	97.06±1.30	98.65±0.98	98.93±00.9	83.19±3.64	87.32±1.10	88.12±8.57
2	67.71±8.74	83.14±10.81	96.43±2.83	84.05±6.56	89.69±7.12	91.71±4.33	82.82±8.74	93.12±6.91	**98.28±1.60**	95.11±0.52
3	93.64±0.51	96.51±1.49	95.17±4.11	**100±0.00**	98.71±1.12	98.61±1.26	97.03±1.01	96.79±1.86	99.52±0.34	98.53±1.08
4	89.09±2.04	89.82±5.69	94.82±2.38	89.97±0.41	92.89±3.83	90.73±5.86	85.46±9.5	90.41±4.94	89.03±6.80	**98.49±0.12**
5	92.61±1.81	94.83±2.06	98.91±0.51	96.25±3.19	95.62±4.02	94.25±3.87	96.87±1.08	96.13±3.01	99.52±0.34	**99.73±0.18**
6	78.73±6.45	90.06±4.87	**97.48±2.48**	91.86±4.78	88.53±6.21	86.75±8.6	90.93±8.88	85.40±3.98	85.33±5.29	87.85±0.84
7	87.07±3.02	94.28±1.58	91.58±3.16	82.32±6.29	85.36±7.97	86.01±4.23	95.82±3.38	92.58±0.85	92.52±1.59	**96.72±0.80**
8	73.08±5.91	98.33±0.87	74.63±4.82	76.84±7.06	80.39±7.37	90.38±6.84	91.70±0.59	97.50±1.27	96.05±0.16	**99.35±0.59**
9	94.41±2.11	**95.26±2.91**	86.75±3.58	85.23±4.69	87.95±4.13	92.24±2.19	95.58±4.11	94.67±2.36	96.34±0.29	91.96±6.44
10	**100±0.00**	92.81±3.20	94.23±3.34	93.43±1.93	93.40±1.96	93.70±2.39	91.24±3.84	91.89±2.6	91.86±1.44	97.92±1.25
11	94.43±2.03	97.85±2.11	94.65±2.73	97.03±1.5	96.17±1.38	95.91±1.03	98.50±1.31	97.85±2.11	**99.68±0.25**	99.28±0.52
12	80.19±2.73	95.93±2.65	89.55±1.93	92.26±2.23	95.02±1.68	96.48±1.97	95.47±2.33	98.17±0.59	**98.70±0.30**	96.90±0.57
13	**99.22±0.55**	94.06±4.59	96.80±3.08	95.62±0.08	93.47±3.00	93.93±3.38	96.33±2.02	93.16±3.47	95.68±2.33	94.04±0.80
14	**100±0.00**	92.38±6.91	**100±0.00**	98.99±0.71	99.50±0.21	**100±0.00**	**100±0.00**	92.30±6.83	98.64±1.29	98.33±1.23
15	97.85±1.17	**99.84±0.13**	98.02±1.42	97.26±1.87	97.21±1.63	96.91±3.03	99.74±0.14	99.74±0.19	98.62±1.31	98.27±0.56
OA	87.28±0.03	92.71±1.13	92.51±0.73	90.24±0.41	91.49±1.37	92.77±1.60	93.10±0.27	93.37±1.49	94.94±0.27	**96.05±0.38**
AA	89.87±0.70	93.48±1.33	93.57±0.60	91.82±0.87	92.73±0.86	93.75±0.68	94.43±0.57	93.53±1.38	95.14±0.23	**96.04±0.42**
Kappa	86.25±0.04	92.12±01.22	91.89±0.78	89.45±0.44	90.81±1.48	92.18±1.73	92.54±0.30	92.83±1.61	94.53±0.29	**95.73±0.41**

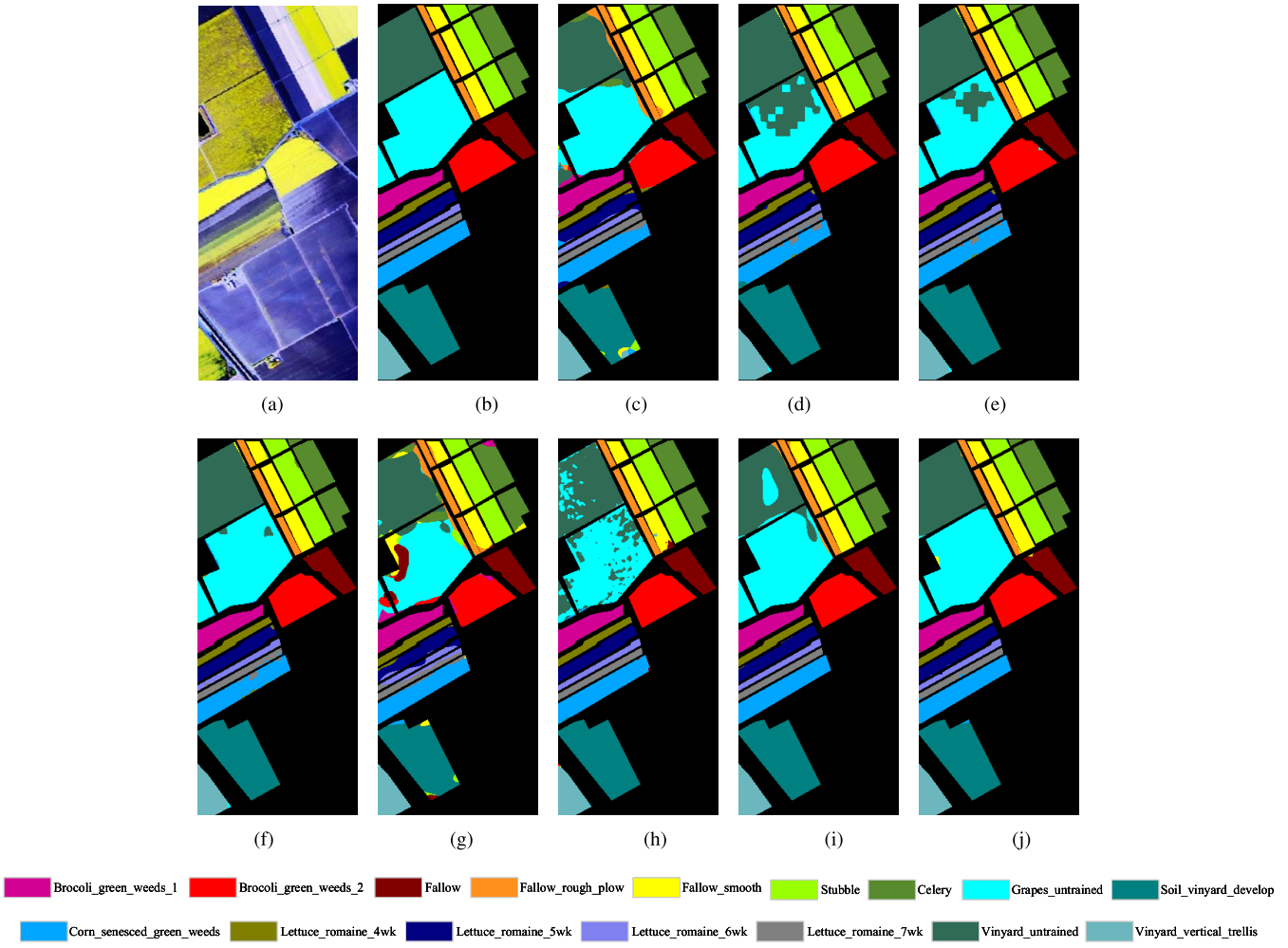


Fig. 8. Visualization results on the Salinas dataset. (a) False Color. (b) GT. (c) Mini-GCN. (d) Nonlocal-GCN. (e) CAD-GCN. (f) MD-GCN. (g) DC-CNN. (h) SSRN. (i) DFCN. (j) Ours.

is the dynamic graph is constantly updated according to the HSI data distribution and HSIC tasks (that is, dynamic graph, denoted as "DG"). To verify the effectiveness of these two modules, ablation experiments have been performed on four datasets. "GCN" represents the experimental setting that the adjacency matrix is calculated using the Euclidean distance, and the graph is fixed during the training; then, the resulting graph representation is fused with deep CNNs' features for classification. "GCN + LA" means that the nodes' similarity metric in "GCN" is replaced from Euclidean distance to SiamNet. "GCN + LA + DG" (i.e., the proposed Auto-GCN) indicates replacing the fixed graph in "GCN + LA" with

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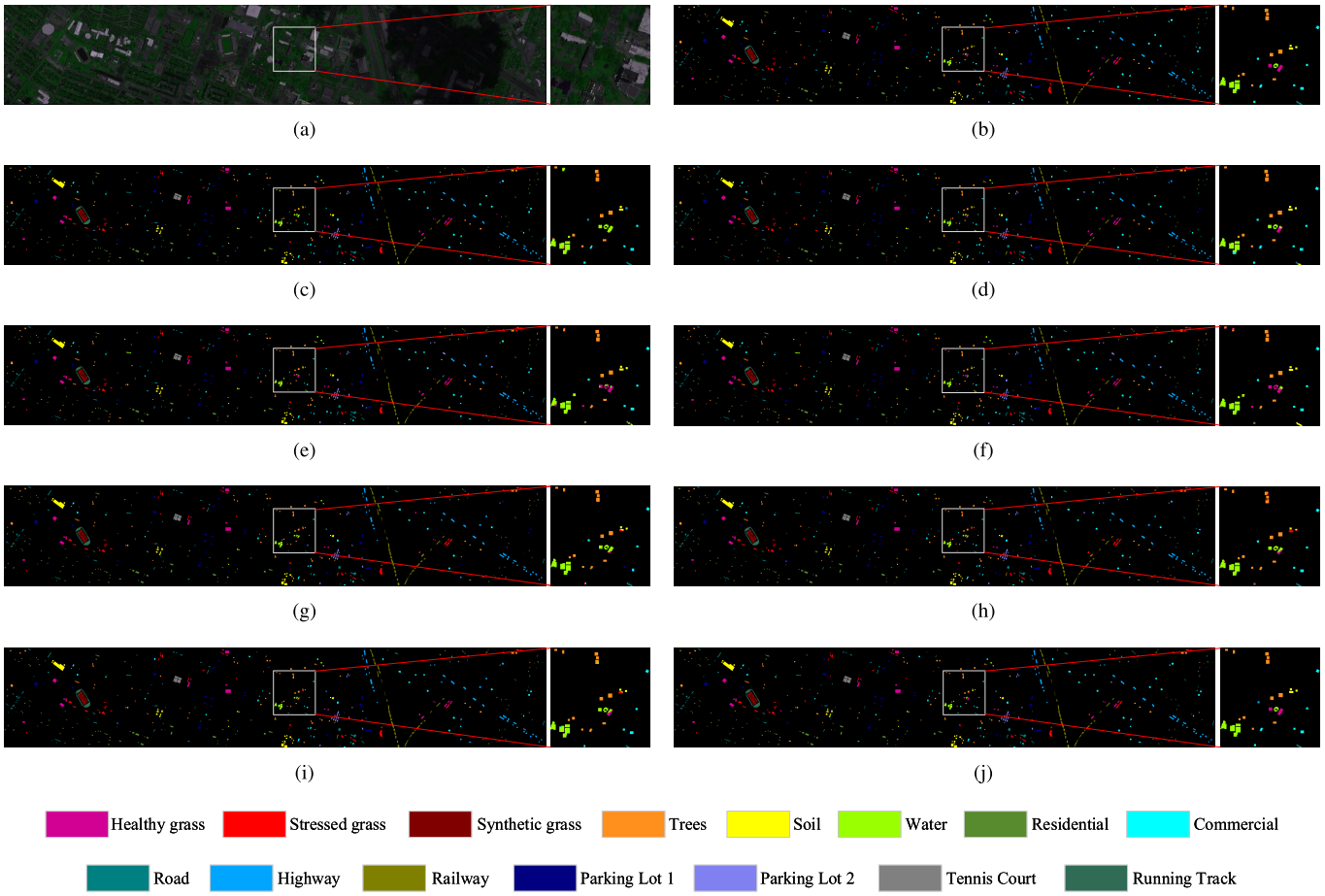


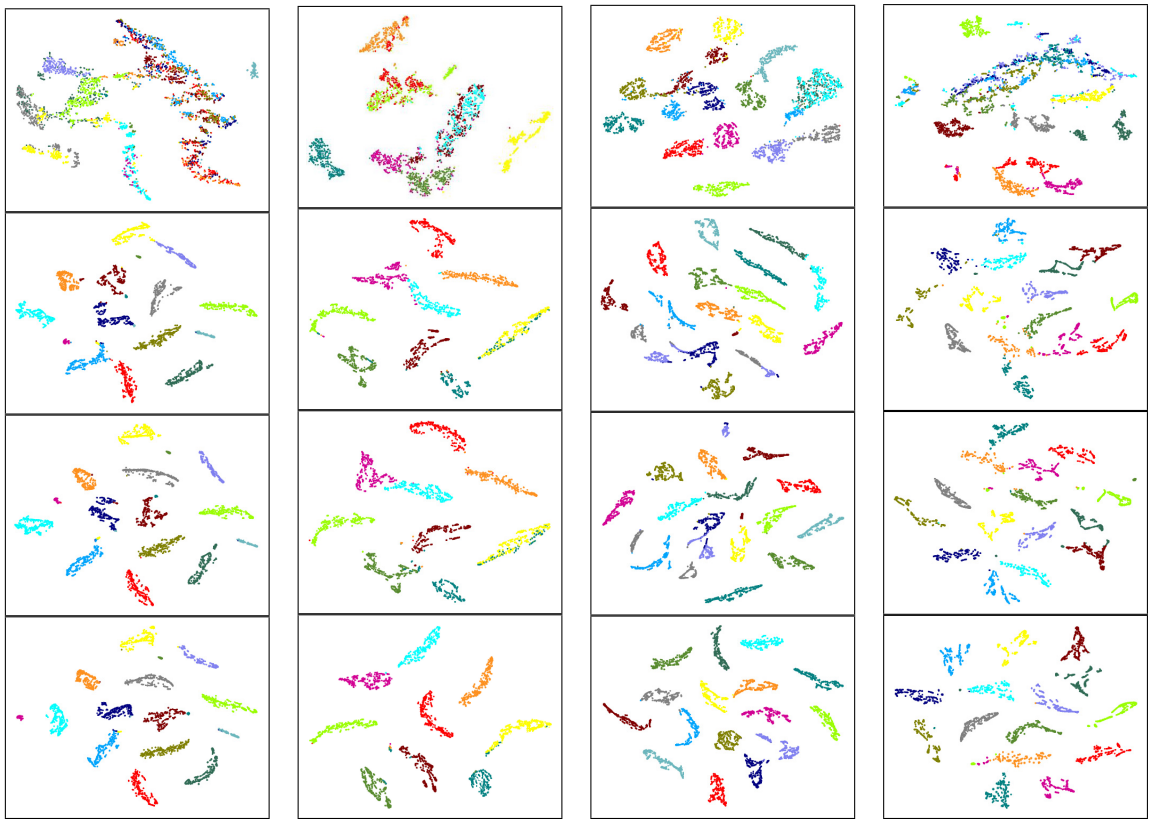
Fig. 9. Visualization results on the Houston University dataset. (a) False Color. (b) GT. (c) Mini-GCN. (d) Nonlocal-GCN. (e) CAD-GCN. (f) MD-GCN. (g) DCNN. (h) SSRN. (i) DFCN. (j) Ours.

the dynamic graph, which changes as the gradient decreases. It can be observed from Tables V–VIII that the classification performance of the proposed method is improving by adding "LA" and "DG" modules. Auto-GCN obtains better results than other algorithms. It demonstrates that computing edges between nodes using Euclidean distances cannot accommodate complex data distributions, and the task-oriented dynamic graph representations are more robust than the fixed graphs.

Figs. 6–9 show the classification maps of the proposed and comparison algorithms on the labeled pixels of the four datasets, respectively. Compared with the superpixel-based GCNs method (MD-GCN and CAD-GCN), the proposed method does not suffer from wide-range misclassification. The reason is that, when computing the adjacency matrix, the graph generated by the superpixel algorithm only considers the neighborhood nodes in a small region, and edges between nodes are calculated by the manually designed distance function, while the neighbor information of our method includes the whole image, and edges are obtained by deep learning. In the compared CNNs-based methods, the classification results of the SSRN algorithm on the Indian Pines and Salinas datasets appear large-scale dotted misclassification, which is due to the fact that the training samples of the SSRN algorithm are taken in small windows around each pixel, causing the lack of spatial context, while the results of DC-CNN and DFCN algorithms improve, which reaffirms the significance of global information.

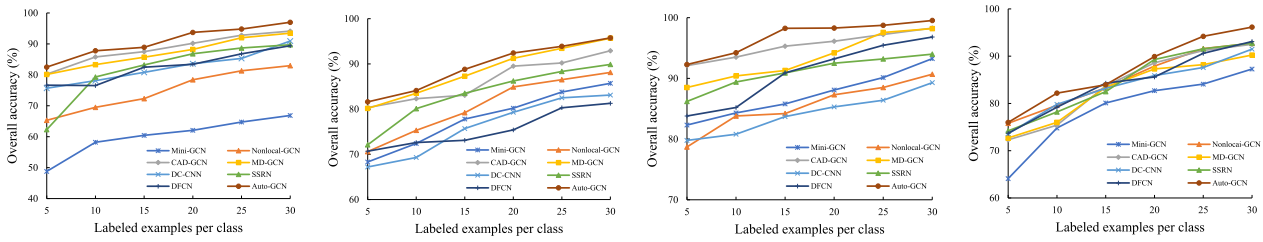
We can see from the visualization that the comparison algorithms all have large regions of color confusion. For example, in the Indian Pines dataset (see Fig. 6), the "Soybean-clean" is easily classified into other categories and leads to an increased misclassification rate. For the Pavia University dataset (see Fig. 7), observing the red domain (denotes "Meadows") in the right bottom corner of each plot, with no interference from the surrounding similar categories, there is patchiness in the visualization figures of algorithms Mini-GCN, Nonlocal-GCN, DC-CNN, SSRN, and DFCN. For the Salinas dataset, the comparison results in Table VII show that our algorithm achieves relatively outstanding results, as can be observed from the visualization results (see Fig. 8) that "Grapes_untrained" is covered by a large portion of "Vinyard_untrained", which causes highly misclassified "Grapes_untrained", while this situation is significantly alleviated by our algorithm. For the Houston University dataset, the results in Table VIII demonstrate that the proposed Auto-GCN still achieves promising results in processing the large-scene hyperspectral dataset. In Fig. 9, we zoom in the white rectangular box and can see that the comparison algorithms all have different degrees of underclassification and misclassification, whereas the classification map of Auto-GCN performs well. The above evidence suggests that our approach encodes the global context into

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(a) (b) (c) (d)

Fig. 10. Feature embedding visualizations of three ablation experiments on four datasets. Different rows correspond to the feature distribution of different algorithms, specifically, the original spectrum (the first row), "GCN" (the second row), "GCN + LA" (the third row), and "GCN + LA + DG" (the fourth row). (a) Indian Pines dataset. (b) Pavia University dataset. (c) Salinas dataset. (d) Houston University dataset.



(a) (b) (c) (d)

Fig. 11. OA of various algorithms in case of different labeled sample numbers per class. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Salinas dataset. (d) Houston University dataset.

data-driven and task-oriented graph representations, and the flowing global information effectively reduces the misclassification rate between similar categories.

E. Visualizing the Features of Ablation Experiments

To qualitatively evaluate the effectiveness of Auto-GCN, we conduct the t-SNE [83] visualization of three sets of ablation experiments on four datasets. To clearly visualize the distribution of features, we randomly select 300 pixels from each category in the test dataset and take all samples if there are less than 300. Fig. 10 shows the feature embeddings of original spectral features, "GCN," "GCN + LA," and "GCN + LA + DG" (i.e., Auto-GCN) on the four datasets, respectively, where different colors indicate different categories of labels. It can be observed that the original spectral feature distributions of all datasets are seriously mixed together, that is, samples of different categories have a spectral similarity. With the addition of the "LA" and "DG" modules, the pixels in the clusters become more compact, and the edges between the clusters gradually increase. Experiments show that graph representations obtained from the learnable adjacency matrix and dynamic graph learning provide more robust classification features for hyperspectral classification, which can effectively reduce intraclass distances and increase interclass distances.

F. Impact of the Number of Training Samples

In this section, we investigate the effect of different numbers of labeled samples on the classification accuracy of Auto-GCN and the seven comparison algorithms. In our experiments, the number of labeled samples per category is set from five to 30 pixels with an interval of 5. Fig. 11 records the overall accuracy (OA) of all methods on four datasets (Indian Pines, Pavia University, Salinas, and Houston University datasets, respectively). As the results show, the proposed Auto-GCN

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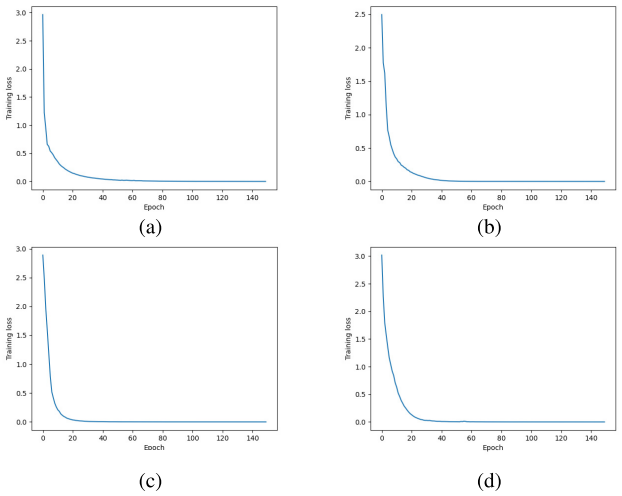


Fig. 12. Training loss curves of Auto-GCN on four datasets. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Salinas dataset. (d) Houston University dataset.

is still competitive under the insufficient labeled samples, and it is particularly evident on the Salinas dataset, which also proves that our method has a strong learning ability for small samples. This is because the off-line training process of the dynamic graph contains abundant prior information of HSI. Moreover, the update of the dynamic graph is driven by the classification task, which helps to enhance the expression of the graph representation.

G. Convergence Analysis

To evaluate the convergence of Auto-GCN, we show its loss curves on the four datasets in Fig. 12. As can be observed from the loss plots of the four datasets, Auto-GCN converges rapidly within 30 epochs, and then, the loss reaches a minimum and plateaus from 30 to 150 epochs. This is due to that we first train SiamNet offline using the relabeled HSI patches in Section III-A, and then, the parameters of SiamNet's feature extractor are employed to initialize the feature extraction network layer of Auto-GCN. Therefore, the initial loss of Auto-GCN is small, and it can be converged to a smooth state rapidly.

H. Computation Time Comparison

Table IX records the running time of five HSIC algorithms, including Mini-GCN, MD-GCN, CAD-GCN, SSRN, and the proposed Auto-GCN on four datasets (i.e., Indian Pines, Pavia University, Salinas, and Houston University datasets). The experimental conditions are the same as Section IV-B. Although our method requires additional computation to evaluate the similarity between nodes via SiamNet, it can be observed that Auto-GCN still shows high efficiency, especially on the Indian Pines and Salinas datasets. It is greatly due to the shared feature extractor between the proposed Auto-GCN and the off-line trained SiamNet. Meanwhile, the region-based graph modeling approach in this article can not only capture the high-dimensional abstract node features but also reduce the graph nodes' number, which significantly accelerates the convergence speed of Auto-GCN.

TABLE IX
RUNNING TIME (S) OF DIFFERENT ALGORITHMS ON FOUR DATASETS

Dataset	Mini-GCN [27]	MD-GCN [33]	CAD-GCN [34]	SSRN [81]	Auto-GCN
IP	65	95	62	201	**60**
PU	240	244	**73**	180	102
Sa	594	1108	826	230	**218**
HU	179	360	**159**	350	312

V. CONCLUSION

In this article, a novel algorithm called Auto-GCN is proposed for HSIC. We perform the automatic learning of graph representations by the "network-in-network" approach. The interaction of higher order tensors is structured by the graph, where nodes are represented by high-dimensional abstract features of the image regions. To make the adjacency matrix better match the data distribution of HSI, we embed a binary semisupervised SiamNet in GCNs to calculate the similarity between high-dimensional nodes. The topological information of the HSI is further encoded and inferred by GCNs, and then, the learnable graph reprojection matrix maps the graph encoding back to grid features to obtain the associative representation of the image. To augment the expressiveness of the graph representation, the subnetwork and the backbone network are trained and updated collaboratively. Thus, the proposed Auto-GCN is both HSI data and HSIC task-driven.

Besides, although Auto-GCN performs well on hyperspectral classification tasks, it is worth noting that, unlike datasets containing multi-images, the existing publicly available hyperspectral data are all one-image only. In the future, we will validate the effectiveness of Auto-GCN on larger datasets and other computer vision tasks.

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Jie Chen received the B.S. degree from Xi'an Technological University, Xi'an, China, in 2017. She is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, School of Artificial Intelligence Xidian University, Xi'an.

Her research interests include deep learning, image processing, and graph representation learning.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. His research interests include image processing, natural computation, machine learning, and intelligent information processing.

Dr. Jiao is also a Foreign Member of the Academia Europaea and the Russian Academy of Natural Sciences, a fellow of the Institution of Engineering and Technology (IET)/Chinese Association for Artificial Intelligence (CAAI)/Chinese Institute of Electronics (CIE)/China Computer Federation (CCF)/Chinese Association of Automation (CAA), a Councilor of CIE, a Committee Member of the Chinese Committee of Neural Networks, and an Expert of the Academic Degrees Committee of the State Council. He is also the Chairperson of the Awards and Recognition Committee. He is also the Vice Board Chairperson of CAAI.

Xu Liu (Member, IEEE) received the B.Sc. degree in mathematics and applied mathematics from the North University of China, Taiyuan, China, in 2013, and the Ph.D. degree from Xidian University, Xi'an, China, in 2019.

He is currently a Post-Doctoral Researcher with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, School of Artificial Intelligence, Xidian University. His research interests include machine learning and image processing.

Dr. Liu was the Chair of the IEEE Xidian University Student Branch from 2015 to 2019.

Lingling Li (Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2011 and 2017, respectively.

From 2013 to 2014, she was an Exchange Ph.D. Student with the Intelligent Systems Group, Department of Computer Science and Artificial Intelligence, University of the Basque Country UPV/EHU, Leioa, Spain. She is currently a Post-Doctoral Researcher with the School of Artificial Intelligence, Xidian University. Her research interests include quantum evolutionary optimization, machine learning, and deep learning.

Fang Liu (Senior Member, IEEE) received the B.S. degree in computer science and technology from Xi'an Jiaotong University, Xi'an, China, in 1984, and the M.S. degree in computer science and technology from Xidian University, Xi'an, in 1995.

She is currently a Professor with the School of Computer Science, Xidian University. Her research interests include signal and image processing, synthetic aperture radar image processing, multiscale geometry analysis, learning theory and algorithms, optimization problems, and data mining.

Shuyuan Yang (Senior Member, IEEE) received the B.A. degree in electrical engineering and the M.S. and Ph.D. degrees in circuits and systems from Xidian University, Xi'an, China, in 2000, 2003, and 2005, respectively.

She has been a Professor of electrical engineering with Xidian University. Her research interests include machine learning and multiscale geometric analysis.

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